

$$\begin{array}{ccc}
 U \subseteq \mathcal{A}_0 & \xrightarrow{f} & \mathcal{A}_0 \\
 \phi \downarrow & & \uparrow \psi \\
 \mathbb{D}_r & \xrightarrow{z \mapsto z^n} & \mathbb{D}_r
 \end{array}$$